

# COMPUTER PROGRAMMING

Electrical Engineering Department — UET Nowshera

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## LAB 1

Lab Title: Installation of DEV-C++ IDE and Execution of First C++ Program

### 1. OBJECTIVE

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The purpose of this lab is to gain hands-on experience with installing the DEV-C++ Integrated Development Environment (IDE) and executing a fundamental C++ program. Students are introduced to the process of writing, compiling, and running a basic C++ program within the DEV-C++ environment. Additionally, students learn how to create a GitHub account and upload their source code files to a remote repository for submission and version control.

### 2. THEORY

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C++ is a powerful, general-purpose programming language extensively used in the development of software and engineering solutions. In order to write, compile, and execute C++ programs, a suitable programming environment called an Integrated Development Environment (IDE) is essential.

DEV-C++ is a beginner-friendly IDE designed for new programmers, offering a unified environment to write, compile, and run programs without switching between separate tools. It includes a source-code editor, an integrated compiler, and a dedicated output window. In C++, program execution always begins from the `main()` function, which acts as the entry point of every program. The `cout` statement is responsible for displaying output on the console screen.

GitHub is a web-based platform used for version control and collaborative software development. It allows programmers to store their source code in remote repositories, track changes over time, and share their work with others. In this lab, GitHub is used to upload and submit the completed C++ source file.

### 3. SOFTWARE USED

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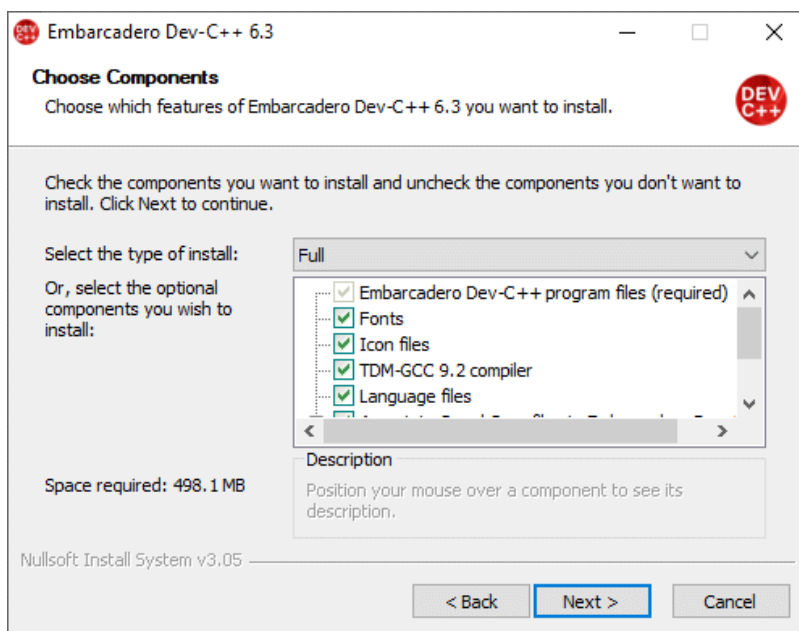
- DEV-C++ IDE
- C++ Compiler (bundled with DEV-C++)
- GitHub Account

## 4. PROCEDURE

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### Step 1 — Install DEV-C++

- Visit the official DEV-C++ website and download the installation package.
- Run the installer and carefully follow each step shown on the screen.
- Once installation is complete, launch the DEV-C++ application.



### Step 2 — Create a New C++ Source File

- Navigate to File → New → Source File to open a new, empty editor window.
- Go to File → Save As and save the file with the name lab1\_hello.cpp.

### Step 3 — Write the First C++ Program

Type the following code into the editor window. The inline comments (beginning with `//`) describe what each line does:

```
// Include the standard input/output stream library (needed for cout)
#include <iostream>

// Allow use of standard library names without the 'std::' prefix
using namespace std;

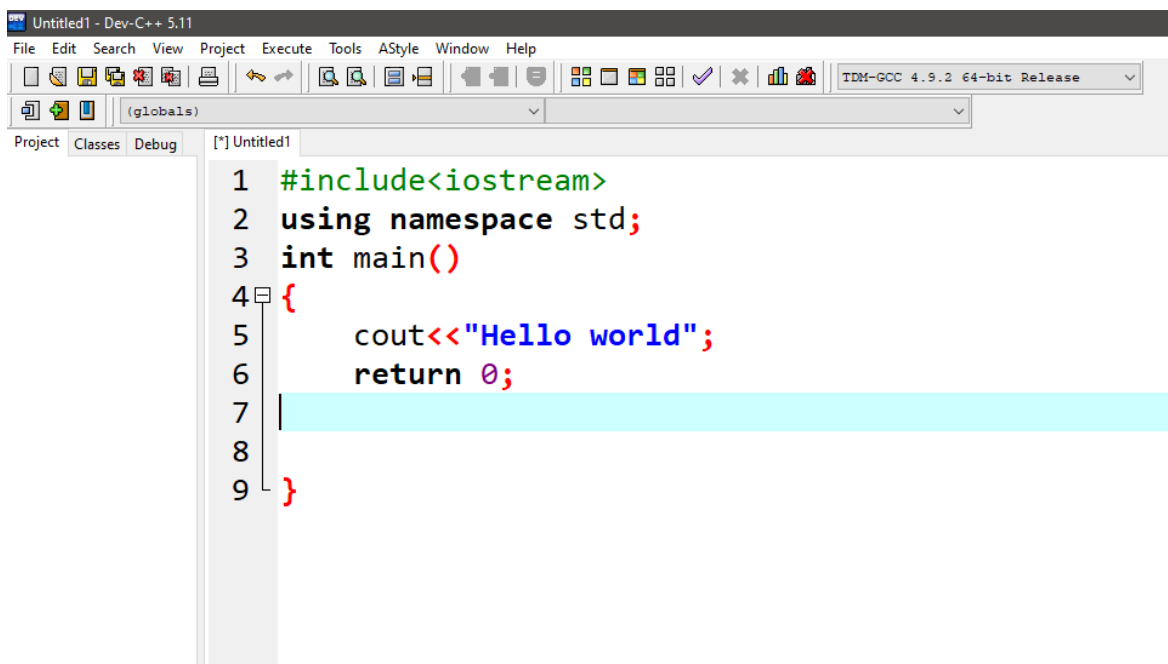
// main() - the entry point of every C++ program
int main()
{
    cout << "Hello World!"; // cout prints text to the console window
    return 0;               // return 0 signals successful program
    termination
}
```

#### Step 4 — Compile and Run the Program

- Click Execute → Compile to build the source code and verify there are no errors.
- Click Execute → Run to launch the compiled program.
- The console output window should display the following result:

#### Output:

Hello World!



## 5. GITHUB TASK

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### Step 1 — Create a GitHub Account

- Open the official GitHub website at github.com.
- Click the Sign Up button.
- Enter your email address, preferred username, and a secure password.
- Verify your account by completing the on-screen prompts.
- Once registration is complete, log in to your GitHub account.

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practicalseries-lab

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lab@practicalseries.com

Password

.....

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Frictionless development

Open source community

By clicking on "Create an account" below, you are agreeing to the [Terms of Service](#) and the [Privacy Policy](#).

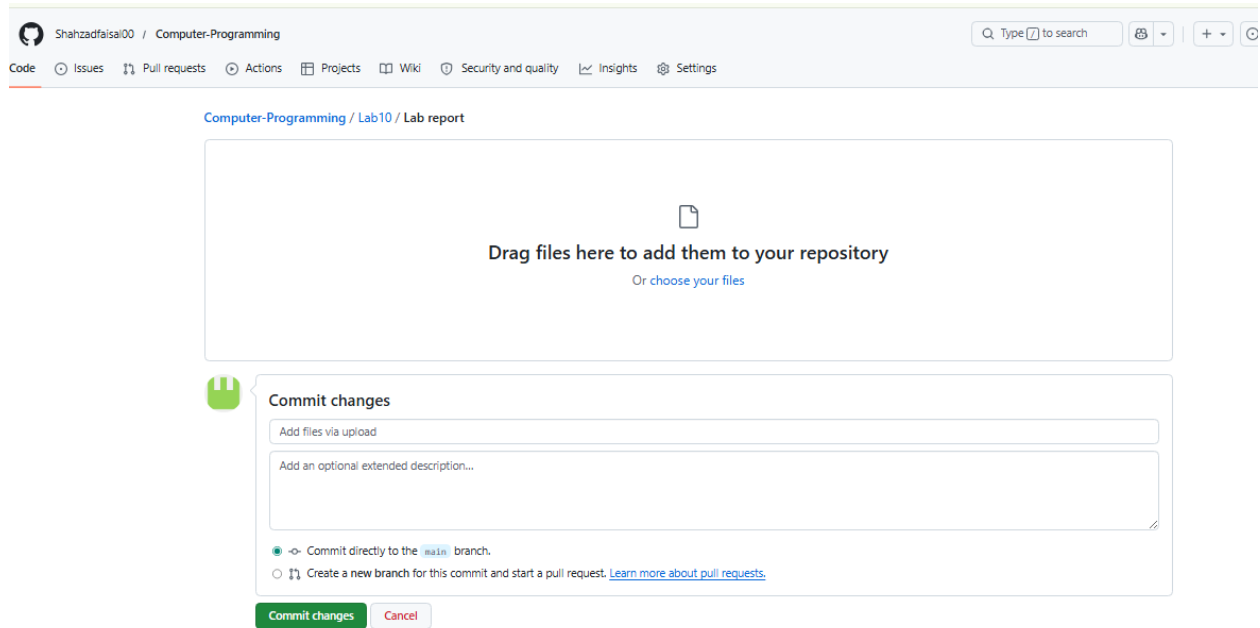
Create an account

### Step 2 — Create a Repository Named “Cpp-Lab-Tasks”

- After logging in, click the + icon in the top-right corner of the page.
- Select New repository from the drop-down menu.
- In the repository name field, type: Cpp-Lab-Tasks.
- Set the repository visibility to Public.
- Click the Create Repository button to confirm.

### Step 3 — Upload the File “lab1\_hello.cpp”

- Open the newly created Cpp-Lab-Tasks repository.
- Click the Add File button and select Upload Files.
- Drag and drop the file lab1\_hello.cpp, or click Choose Your Files to locate it on your computer.
- After the file has been uploaded, click the Commit Changes button to save it to the repository.



The GitHub repository stores lab files securely online and enables backup, version control, and easy sharing of programming work. The repository link submitted to the instructor for this lab is:

GitHub Repository Link: <https://github.com/talha67ali/Computer-Programming-Lab>

## 6. CONCLUSION

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This lab provided a practical introduction to C++ programming using the DEV-C++ IDE. The program was successfully written, compiled, and executed, producing the expected output ‘Hello World!’ on the console. Through this exercise, key C++ constructs were understood: the role of `#include <iostream>` for enabling input and output operations, using namespace `std` for simplified

syntax, the `main()` function as the program entry point, `cout` for displaying output, and `return 0` as a standard termination signal. In addition, the GitHub workflow was practised, including account creation, repository setup, and source file upload, establishing a foundation for version-controlled lab submissions throughout the course.